

AMENDMENT NO. 2 JUNE 2026
TO
IS 16647 : 2017 ORIENTED UNPLASTICIZED POLYVINYL CHLORIDE (PVC-O)
PIPES FOR WATER SUPPLY — SPECIFICATION

(First cover page, title) — Substitute the following for the existing:

‘जल आपूर्ति और सीवेज हेतु उन्मुख अनप्लास्टिस्टाइज्ड पॉलीविनाइल क्लोराइड (पीवीसी-ओ) पाइप — विशिष्टि

**Oriented Unplasticized Polyvinyl Chloride (PVC-O) Pipes for Water Supply
and Sewage — Specification’**

(Foreword, second para, last sentence) — Substitute the following for the existing:

‘The PVC-O pipes covered under this standard are primarily intended for the conveyance of water under pressure in water supply systems. In addition to their established use in water supply, PVC-O pipes are also suitable for sewage, wastewater, and buried drainage applications, including the conveyance of treated wastewater, owing to their strength, durability, and resistance to corrosion and chemical attack, subject to appropriate design and installation conditions. This standard applies to PVC-O pipes intended for use in buried or otherwise protected conditions and not for applications involving direct exposure to sunlight.’

(Foreword, last para, line 2) — Substitute ‘IS 2 : 2022 Rules for rounding off numerical values *(second revision)*’ for ‘IS 2 : 1960 Rules for rounding off numerical values *(revised)*’.

(Page 1, title) — Substitute the following for the existing:

**‘ORIENTED UNPLASTICIZED POLYVINYL CHLORIDE (PVC-O) PIPES FOR WATER SUPPLY
AND SEWAGE — SPECIFICATION’**

[Page 1, clause 1.1, (see also Amendment No. 1)] — Substitute the following for the existing:

‘This standard specifies the requirements of oriented unplasticized polyvinyl chloride (PVC- O) Pipes, for piping systems intended to be used underground, or above ground but not exposed to direct sunlight, for water supply, buried drainage, sewage, and treated wastewater.’

(Page 1, clause 1.2) — Substitute the following for the existing:

‘The piping system according to this standard is intended for the conveyance of cold water under pressure and is suitable for the conveyance of water, including potable water, sewage, and treated wastewater, up to and including 45 °C, and especially in applications requiring special performance characteristics, such as resistance to impact loads and pressure fluctuations, up to a pressure of 2.5 MPa.’

(Page 3, clause 5.2) — Substitute the following for the existing:

‘The addition of the manufacturer’s own rework material is permissible. The maximum quantity of the rework material used shall be not more than 5 percent. No other rework material shall be used.’

(Page 3, clause 5.6.1) — Insert the following at the end:

‘In order to establish the validity of classification of MRS value by the manufacturer and to get qualified for the production of respective MRS class of PVC-O pipes, an additional type approval test for long-term hydrostatic strength at 27 °C for 10 000 h shall be carried out once in accordance with 9.1.1. Such test shall also be carried out whenever any change in the manufacturing process occurs. Alternatively, such test carried out by the technology provider from any accredited laboratory shall also be accepted’.

Price Group 2

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(Page 7, clause 9.1.1, line 4) — Substitute ‘10 h at 27 °C, 1 000 h at 27 °C and 10 000 h at 27 °C’ for ‘10 h at 27 °C and 1 000 h at 27 °C’.

(Page 7, clause 9.1.1, para 2) — Insert the following new para at the end:

‘For a period of 10 000 h, values established from analysis of test data in accordance with IS 16462/ISO 9080 can be taken as minimum stress level.’

[Page 8, table 10, (see also Amendment No. 1)] — Insert the following at the end:

SI No.	Test	Temperature, <i>Min</i> °C	Duration, <i>Min</i>	Circumferential Stress/Hoop Stress, σ	
(1)	(2)	(3)	(4)	(5)	(6)
iii)	Type [one time type approval test (<i>see</i> 5.6.1)]	27	10 000	45	50

[Page 9, table 12, footnote (2)] — Substitute the following for the existing:

²⁾ Due to specific jointing techniques used in water supply, sewage, and treated wastewater systems, as well as installation below ground, higher stiffness may be required.’

[Page 10, table 14, SI No. (i), col (2)] — Substitute the following for the existing:

‘T = Temperature within ± 5 °C of any temperature between 20 °C and 32 °C’

[Page 10, table 15, SI No. (i), col (2)] — Substitute the following for the existing:

‘T = Temperature within ± 5 °C of any temperature between 20 °C and 32 °C’

(Page 11, clause 12) — Substitute the following for the existing:

‘12 SEALING RINGS

Sealing rings made of elastomeric material such as ethylene propylene diene monomer (EPDM) rubber, and of appropriate diameter ensuring secure fit shall be used for joining components and shall meet the following requirements:

- Rings shall conform to the material requirements specified in IS 5382 and shall have shore hardness class 70 as per IS 5382. Also, the manufacturer has to specify the application type of sealing ring that is being offered (see IS 5382). The design of the profile of the sealing ring is left to the manufacturer as long as the pipe with sealing ring meets the requirements of the standard; and
- Rings shall be free from chemical agents such as plasticizers that could have a detrimental effect on the pipes or fittings, or on the quality of the water.

A test report or conformity certificate may be obtained from the manufacturer of the elastomeric sealing ring for the conformity to IS 5382. The frequency of this test report or conformity certificate shall be once in six months.’

(Page 11, clause 14.1, line 3) — Substitute ‘not more than 3 m’ for ‘not more than 1 m’.

(Page 12, Annex A) — Substitute the following entries for the existing:

IS No.	Title
IS 4985 : 2021	Unplasticized PVC pipes for water supplies — Specification (<i>fourth revision</i>)
IS 5382 : 2025/ISO 4633 : 2023	Rubber seals — Joint rings for water supply, drainage and sewerage pipelines — Specification for materials (<i>third revision</i>)
IS 10148 : 2023	Positive list of constituents of polyvinyl chloride (PVC) and its copolymers in contact with foodstuffs, pharmaceuticals and drinking water (<i>first revision</i>)

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<i>IS No.</i>	<i>Title</i>
IS 10151 : 2019	Polyvinyl chloride (PVC) and its copolymers for its safe use in contact with foodstuffs, pharmaceuticals and drinking water — Specification (<i>first revision</i>)

(Page 18, clause **G-1.9**, line 7) — Substitute ‘Table 25 or Table 26’ for ‘Table 24 or Table 25’.